

US Domestic Passenger Traffic

2021-2025 | Vaibhav Soni | SQL · Google BigQuery Sandbox

> ISSUE / PROBLEM

After growth through 2024, scheduled segment passengers declined in 2025 while offered capacity increased.

The project asks **where demand and capacity diverged** across years, months, reporting carriers and airports.

> RESPONSE

Loaded and profiled **five annual BTS extracts**: 2,107,690 raw rows.

Built a reusable SQL view of **1,579,887 valid scheduled-service records (Class F)**, covering all 12 months in each year.

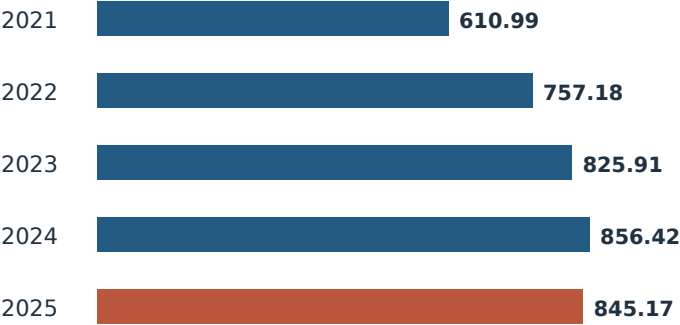
Answered six questions using demand, seat supply, departures and distance-weighted load factor.

> IMPACT / ANALYTICAL VALUE

Supports **carrier, route and month reviews** before service changes. Fare, schedule and cost data are needed to explain causes. No operational or financial impact was measured.

> FIVE-YEAR DEMAND

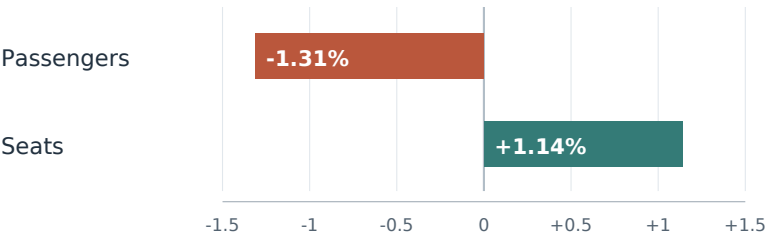
Scheduled segment passengers (millions)



845.17M passengers in 2025: down approximately **11.25M** from the 2024 peak.

> 2025: DEMAND VS CAPACITY

Year-over-year change versus 2024 (%)



Distance-weighted load factor:
83.71% in 2024 → **81.70%** in 2025
A decline of **2.01 percentage points**.

> KEY INSIGHTS

- Monthly weakness:** nine of 12 months fell versus 2024. February had the largest relative decline (**-4.77%**); October increased **1.71%**.
- Carrier shifts:** Spirit recorded the largest passenger loss (**-11.21M**), while United recorded the largest gain (**+3.61M**). Investigate route and month contributions before attributing causes.
*Reporting carriers include regional operators such as SkyWest and Endeavor.
- Network context:** LAX → JFK led directional routes (**1.46M passengers**); Atlanta led departing segment passengers (**44.23M**); Southwest supplied the most seats (**223.17M**).

Boundaries: monthly segment aggregates represent boardings, not unique travelers. Carriers are reporting operators; routes are directional. The data do not establish causes, profitability or a case for adding capacity.